

ldapsearch non-offensive tool to query LDAP servers, <https://docs.ldap.com/ldap-sdk/docs/tool-usages/ldapsearch.html>

Nessus network vulnerability scanner (commercial), <https://www.tenable.com/products/nessus>

netexec (nxc) multi-tool for attacking AD networks over multiple protocols such as SMB, LDAP, WMI; formerly known as *crackmapexec (cme)*, <https://www.netexec.wiki/>

nmap general-purpose network and service scanner, extendable with user scripts, <https://nmap.org/>

nikto web server vulnerability scanner, <https://github.com/sullo/nikto>

OpenVAS network vulnerability scanner, <https://www.openvas.org/>

PowerMad enrolls new virtual computers into an AD, <https://github.com/Kevin-Robertson/Powermad>

PowerUp automatic windows privilege-escalation tool, <https://github.com/PowerShellMafia/PowerSploit/blob/master/Privesc/PowerUp.ps1>

PowerUpSQL automatic Microsoft SQL Server privilege escalation tool, <https://github.com/NetSPI/PowerUpSQL>

PowerView active directory enumeration tool, <https://github.com/PowerShellMafia/PowerSploit/blob/master/Recon/PowerView.ps1>

responder Responder is a network-protocol poisoner, with many built-in server implementation, typically used to force clients to expose credentials or to perform Attacker-in-the-Middle attacks, <https://github.com/lgandx/Responder>

Rubeus windows-based tool used for kerberos attacks, <https://github.com/GhostPack/Rubeus>

rpcclient non-offensive tool used to access Microsoft DCE RPC services, <https://www.samba.org/samba/docs/4.17/man-html/rpcclient.1.html>

SharpView reimplement of PowerView in C#, <https://github.com/tevora-threat/SharpView>

smbclient non-offensive tool used to access Microsoft SMB network shares, <https://www.samba.org/samba/docs/current/man-html/smbclient.1.html>

smbmap allows users to enumerate samba share drives across an entire domain, <https://github.com/ShawnDEvans/smbmap>

social engineer toolkit (set) open-source penetration testing framework focused upon social engineering, <https://github.com/trustedsec/social-engineer-toolkit>

tcpdump network sniffing tool, <https://www.tcpdump.org/>

tshark network sniffing tool, <https://linux.die.net/man/1/tshark>

Received 16 February 2025; revised 21 August 2025; accepted 24 August 2025